

Matplotlib Cheat Code

ML-focused quick reference — essential plotting only

1. Import

```
import matplotlib.pyplot as plt
```

2. Bar — compare categories

```
plt.bar(df["Product"], df["Revenue"])
plt.title("Revenue by Product")
plt.xlabel("Product")
plt.ylabel("Revenue")
plt.xticks(rotation=45)
plt.show()
```

CATEGORY → COMPARISON

3. Line — trends / ordered data

```
plt.plot(days, sales, marker="o")
plt.title("Daily Sales")
plt.xlabel("Day")
plt.ylabel("Sales")
plt.show()
```

TREND / TIME → LINE

4. Scatter — relationship between numbers

```
plt.scatter(df["Price"], df["Quantity"])
plt.title("Price vs Quantity")
plt.xlabel("Price")
plt.ylabel("Quantity")
plt.show()
```

NUMBER vs NUMBER → SCATTER

5. Histogram — distribution

```
plt.hist(df["Price"])
plt.title("Distribution of Prices")
plt.xlabel("Price")
plt.ylabel("Frequency")
plt.show()
```

DISTRIBUTION → HISTOGRAM

6. Box plot — spread + outliers

```
plt.boxplot(df["Price"])
plt.title("Price Distribution")
plt.ylabel("Price")
plt.show()
```

SPREAD / OUTLIERS → BOX PLOT

7. Pie — parts of a whole

```
category_revenue = df.groupby("Category")["Revenue"].sum()

plt.pie(category_revenue,
labels=category_revenue.index,
autopct="%1.1f%%")
plt.title("Revenue by Category")
plt.show()
```

PARTS OF WHOLE → PIE

8. Figure size

```
plt.figure(figsize=(12, 6))
plt.bar(df["Product"], df["Revenue"])
plt.show()
```

9. Value annotations

```
bars = plt.bar(df["Product"], df["Revenue"])
for bar in bars:
    plt.text(bar.get_x() + bar.get_width() / 2,
             bar.get_height(),
             str(bar.get_height()),
             ha="center", va="bottom")
plt.show()
```

10. Subplots

```
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.bar(df["Product"], df["Revenue"])
plt.title("Revenue by Product")

plt.subplot(1, 2, 2)
plt.plot(days, sales, marker="o")
plt.title("Daily Sales")

plt.tight_layout()
plt.show()
```

`subplot(rows, columns, position)` → `subplot(1, 2, 1)` = 1 row, 2 columns, chart 1.

11. Save a chart

```
plt.savefig("revenue_chart.png")
plt.show()
```

Save before `show()`.

Chart Decision Cheat Code

Question / data	Use
Compare categories	Bar
Trend over time / ordered sequence	Line
Two numerical variables	Scatter
Numerical distribution	Histogram
Spread / possible outliers	Box plot
Parts of a whole	Pie
Multiple charts together	Subplots

ML tip: Focus on choosing the right chart and interpreting it. Advanced styling can be learned when a project needs it.